



Service Tool Instruction

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Description

Camshaft Bushing Replacer Kit

Purpose

This document provides information for using the camshaft bushing replacer kit, Part Number 3165045. The kit is used to remove and install the camshaft bushings on 4B, B3.9, 6B, B5.9, ISB (4 cylinder), ISB^e (4 and 6 cylinder), QSB5.9-44, ISC, QSC8.3, G5.9, B5.9G, B5.9LPG, B Gas Plus, B LPG Plus, C8.3G, C Gas Plus, 6C, C8.3, ISL, QSL9, G3.9, B4.5^s, B6.7^s, QSB3.9-30, QSB4.5-30, QSB5.9-30 series engines.

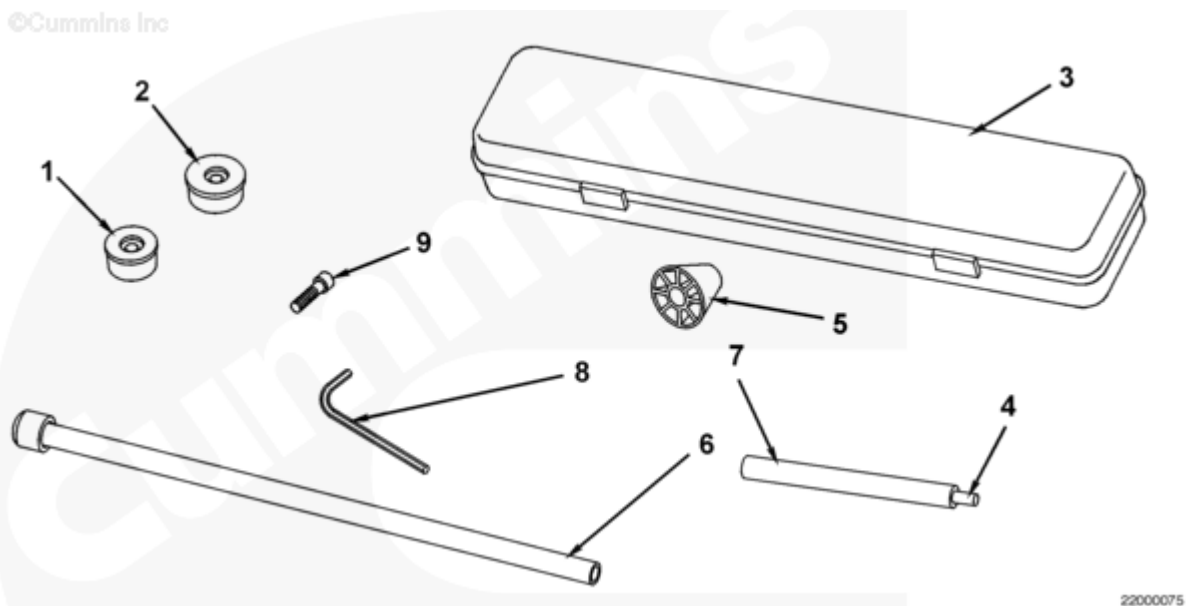


Table 1, Camshaft Bushing Replacer Kit, Part Number 3165045

Item Number	Part Number	Description	Quantity
1	3165038	Camshaft bushing replacer (B-series)	1
2	3165040	Camshaft bushing replacer (C-series)	1
3	3165044	Storage case	1
4	3165042	Socket set screw	1
5	3165063	Centering guide	1

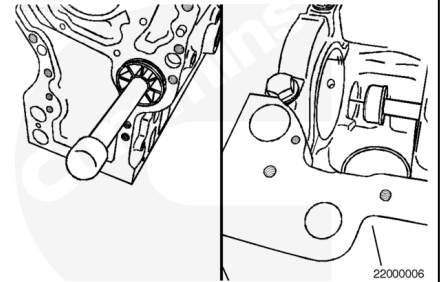
Table 1, Camshaft Bushing Replacer Kit, Part Number 3165045

Item Number	Part Number	Description	Quantity
6	3165064	Drive bar	1
7	3165065	Drive bar extension	1
8	3165067	Hex wrench	1
9	3165068	Socket head capcrew	1

Review the orientation of the oil holes prior to bushing removal for assembly information.

Remove the camshaft bushings using the camshaft bushing replacer kit, Part Number 3165045. Slide the drive bar through the centering guide and insert the drive bar from the rear of the cylinder block through the camshaft bores.

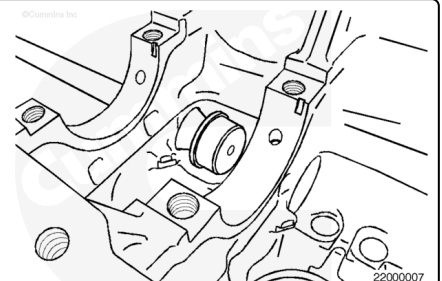
Install the appropriate camshaft bushing replacer and drive the front camshaft bushing from the cylinder block. Remove the replacer and remove the drive bar from the cylinder block.



Note : For engines with a single bushing proceed to the next step.

Insert the drive bar, camshaft bushing replacer, and centering guide from the front of the cylinder block. Drive the remaining camshaft bushings from the cylinder block in succession, starting with camshaft bushing number two.

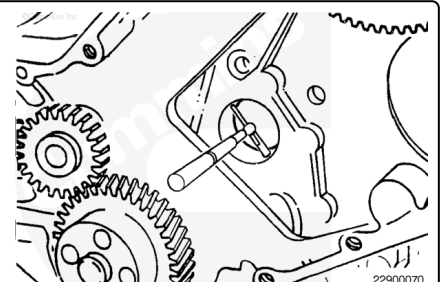
Remove the drive bar, replacer, and guide from the front of the cylinder block.



⚠ WARNING ⚠

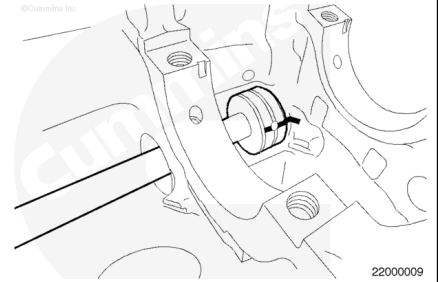
When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

Clean and inspect the cylinder block camshaft bore for excessive wear. Refer to the appropriate Troubleshooting and Repair Manual.



Note : Some camshaft bushings contain multiple oil holes. Make certain the orientation of these holes match the orientation prior to disassembly.

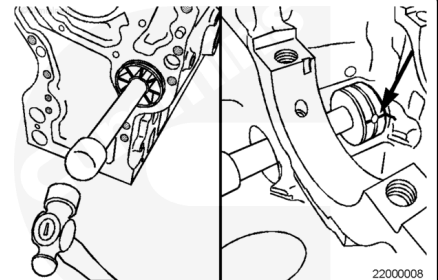
Mark the camshaft bushing and cylinder block using a felt pen. This will align the lubricating oil hole in the cylinder block to the oil hole in the camshaft bushing.



Use the drive bar, centering guide, and appropriate camshaft bushing replacer. Slide the camshaft bushing on the replacer. Align the marks on the camshaft bushing and the cylinder block.

Drive the front camshaft bushing to the correct installed depth. The correct installed depth is when the camshaft bushing oil hole aligns with the cylinder block oil hole.

Install the remaining camshaft bushings in succession, starting at the rear of the cylinder block and working toward the front. Remove the drive bar, replacer, and guide from the cylinder block.



⚠ CAUTION ⚠

To reduce the possibility of engine damage, all camshaft bushings must be checked for proper oil hole alignment after installation.

Make certain that the oil holes are aligned.

A 3.2-mm [0.126-in] diameter rod **must** pass freely through the lubricating oil hole.

Measure the camshaft bushing inside diameter. Refer to the appropriate service manual for specifications.

